Self Driving Vehicles

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Automation Level Chart

SAE (J3016) Automation Levels[82] [hide]

SAE Level	Name	Narrative definition		Execution of steering and acceleration/ deceleration	Monitoring of driving environment	Fallback performance of dynamic driving task	System capability (driving modes)
Human driver monitors the driving environment							
0	No Automation	The full-time performance by the human driver of all aspects of the dynamic driving task, even when "enhanced by warning or intervention systems"		Human driver	Human driver	Human driver	N/a
1	Driver Assistance	The driving mode-specific execution by a driver assistance system of *either steering or acceleration/deceleration*	Using information about the driving environment and with the expectation that the human driver performs all remaining aspects of the dynamic driving task	Human driver and system			Some driving modes
2	Partial Automation	The driving mode-specific execution by one or more driver assistance systems of *both steering and acceleration/deceleration*		System			
Automated driving system monitors the driving environment							
3	Conditional Automation	The driving mode-specific performance by an automated driving system of all aspects of the dynamic driving task	With the expectation that the *human driver will respond appropriately to a request to intervene*	System	System	Human driver	Some driving modes
4	High Automation		*Even if a human driver does not respond appropriately to a request to intervene* the car can pull over safely by guiding system			System	Many driving modes
5	Full Automation		*Under all roadway and environmental conditions* that can be managed by a human driver				All driving modes

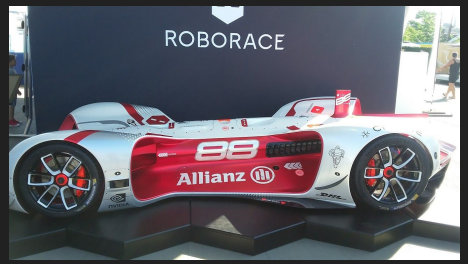
Current Tech

Tesla “Full Self-Driving” - Level 2

Waymo - Level 4

Daimler Truck - Level 2

Roborace - Level 5



Future Tech

Full Release of Level 2 Vehicles

SAE Level 3-5 Vehicles

Cooperative Networking

Watercraft

Aircraft

Current Issues

Tesla Crashes

Responsibility

Driving Profiles

Labels

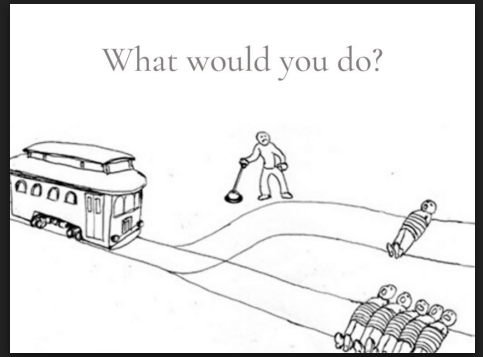
System Confidence Debate

Potential Ethical Dilemmas

Responsibility

Trolley Dilemma

Passenger Variation
- Premium Users?

Cyber Security / Hacking

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